

CONTRIBUTED PAPER

Ten lessons for rare wildlife species conservation when coming from a common species background

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Abstract

Efforts to slow biodiversity loss have ushered in an era of proactive conservation for rare species, or species that have any combination of low relative abundance, high habitat specificity, and limited geographic range. Approaches to wildlife management and conservation were often developed for more abundant, generalist species, and frequently do not translate well to rare species. Conservation for rare species therefore requires adapting knowledge, action and inquiry tools for common species to rare species. We therefore present 10 lessons for adapting conservation tools to rare species, each with advice to implement the lessons. Several themes emerge from these lessons. Information scarcity necessitates high standards of information for rare species. Missing rare species information can be complemented with multiple forms of knowledge, particularly when the need for conservation action precludes additional data collection. Investment in depth (over breadth) of information for individuals of rare species can provide alternatives to population trend, which has lower reliability for rare species. Natural history documentation can improve rare species conservation strategies. Finally, curiosity-driven inquiry can inspire rare species conservation innovation. We hope providing guidance that addresses challenges for rare species will help others bridge the gap between common wildlife approaches and proactive rare species conservation.

KEYWORDS

action, endangered species, inquiry, knowledge, listed species, threatened species

1 | INTRODUCTION

People value rare wildlife species for many reasons, including recognition of intrinsic value, cultural value, and ecosystem services (Angulo & Courchamp, 2009). As a result, there are multiple instances that require societies to use existing knowledge, gain new knowledge, or take action to aid in the preservation of rare species. These include, but are not limited to, the adoption of an

international strategy to address the biodiversity extinction crisis (Convention on Biological Diversity, 2022), shifts in North American wildlife management from a central focus on consumptive use to one where a wider array of stakeholders and species are considered relevant (Feldpausch-Parker et al., 2017; Teel et al., 2022), and formal policy protections through protected designations, such as threatened or endangered status, designated at global (e.g., International Union for the Conservation of

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Nature [IUCN]), national (e.g., United States of America, Canadian, or Australian endangered species acts [ESAs]), and local scales (e.g., state, provincial). Increased interaction with rare species information is also reflected in scientific literature, where there has been a steady increase in rare species science over the past six decades (Figure 1; Golding, 2022a). Unfortunately, much of this activity is hampered not only by logistical challenges of working with rare organisms, but by more fundamentally because much of the research, management, and societal relationships to wildlife conservation have emerged from practices that work best for common species.

Understanding the circumstances where common wildlife species approaches are likely to be less effective for rare wildlife species, and how they can be adapted to support management of rare species, will help broaden our capacity to work toward the multiple forms of rare species conservation goals. Here, we define rare wildlife species as those species characterized by low relative abundance, narrow habitat specificity, a restricted geographic range, or any combination of those features (Crisfield et al., 2024; Gaston, 1994; Rabinowitz, 1981), and note that rarity usually results in some form of data deficiency (Borgelt et al., 2022). This situation contrasts with common species, which boast greater numbers relative to others within a taxonomic group (Gaston & Fuller, 2008), broader distribution (Gaston et al., 2000) and are

increasingly more generalist (Le Viol et al., 2012). A broadening of the species and concepts covered in conservation are not without precedent; in fact, a similar broadening of definitions and capacity occurred in the 1930s when the foundational ideas of North American game management for common harvested species (“game” species) broadened to include a more expanded group of wildlife and became known as “wildlife management” (Brulle & Benford, 2012).

Unfortunately, common species conservation and science approaches have been shown to perform poorly for rare species (Jeliaskov et al., 2022). This underperformance is often driven by the greater importance of an individual of a rare species; individuals matter more to the population, the variation an individual represents has greater influence on information about the species, and these combine to restrict potential methods for information acquisition (e.g., constraints on capture due to concerns about impacts on individuals). In addition, when population monitoring, a ubiquitous tool for understanding common wildlife species, is prioritized over conservation action (Meretsky et al., 2012), there are poorer rare species’ conservation outcomes (Buxton et al., 2020). In management, rarer species often require entirely different plans than existing wildlife conservation plans, such as the Threatened Species Strategy Action Plan in Australia that was created to provide specific targets and resources

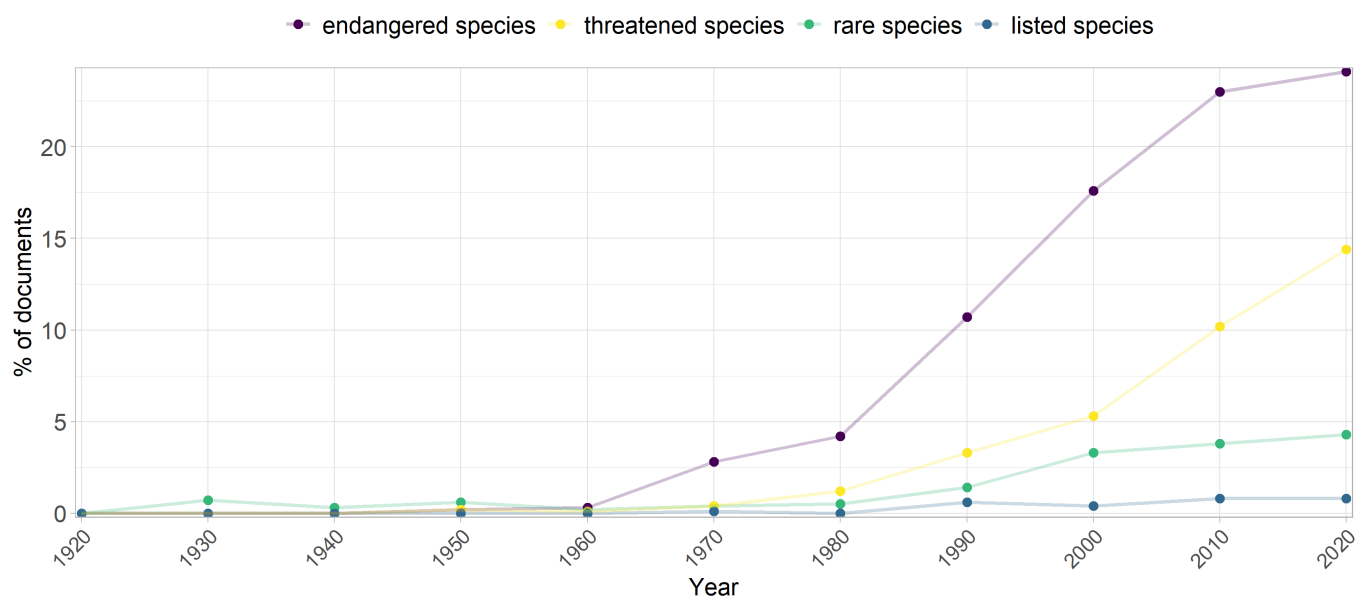


FIGURE 1 The percent of the Journal of Wildlife Management’s publication texts that contain the terms rare species, endangered species, threatened species, or listed species, by decade. We examined all research articles published within JWM between 1937 and 2022 using JSTOR’s text analysis software, Constellate. We selected 4 two-word terms that represented the range of ways in which rare species were discussed both before and after the passage of the US ESA: rare species, endangered species, threatened species, and listed species. We examined all published formats within JWM, including research articles, front matter, book reviews, opinions, and notes, which resulted in the examination of 17,447 different texts, to determine if the text contained any of those terms. Since the 1960s, trends in all four terms have shown a consistent increase by decade, with the highest percentage of texts (24.1%) containing the term endangered species.

for threatened species that have unique challenges not addressed in existing conservation approaches that serve a broad swath of species (Australian Government Department of Agriculture, Fisheries, and Forestry, 2021). Management for more common and generalist species can utilize a wide variety of strategies to improve habitat or affect vital rates, while management for a rare species often requires species- and place-specific measures (e.g., remove a specific non-native predator from a specific site). A greater breadth in potential actions and larger, generalist populations mean the space for success of a common species management or conservation effort is often large, whereas the space for success for a rare species is often narrower.

To that end, and inspired by efforts such as those of Thompson (2004) and Gaston (1994) to distinguish lessons about rarity from those developed primarily for common species for conservation, we aim to provide an updated set of recommendations to incorporate scientific advances we feel will benefit rare species from the past 20 years. We draw on our combined experience of over 50 years as applied wildlife scientists working with rare species primarily in North America, as well as a workshop on this topic put together by Drs. Specht and Golding for the 2019 Annual Conference of the Wildlife Society, where demand for such recommendations was evident. As scientists, educators, and collaborators in applied settings, we have observed burgeoning developments in applied science relevant to rare species management that are adding new strategies and providing insight into the limits of existing tools or approaches, often in ways that have challenged our own or others' intuitions. In many cases, these developments are lessons we wish we had known when we began work on rare species as applied scientists. We classify these developments into categories that we think often encapsulate the fundamental processes for rare species conservation; utilizing knowledge, engaging in inquiry, and taking action. Knowledge, or information that has been previously acquired, forms the foundation for wildlife conservation and further inquiry and action. Unfortunately, we often lack knowledge for environmental decisions at the necessary spatial or temporal scales that would be ideal for action (e.g., Handmaker et al., 2021). We distinguish knowledge from inquiry, or the active process of acquiring new knowledge. Inquiry about rare species has historically been limited to small-scale natural history studies, as it often requires intensive efforts that may not have immediate application. We define action broadly as any activities undertaken to affect a rare species population. We hope that by articulating updated set of lessons for rare species that address knowledge, inquiry, and action challenges for rare species we can help to bridge the gap between existing wildlife conservation

approaches and the growing effort to proactively conserve rare species. Although we present lessons that apply to a single, multiple, or all categories of knowledge, inquiry, or action, we focus primarily on lessons that touch on knowledge and inquiry because these often are the precursors to taking action for rare species, typically because little is known about rare species.

We present these lessons organized by their relevance to knowledge, action, or inquiry (Figure 2) to help readers find the information that is most useful to them. With each lesson, we provide advice to help put the lessons into practice. In some cases, advice provided in different lessons may appear to conflict (e.g., in one lesson we stress the importance of high-quality information for rare species and in another suggest that lack of information should not limit action). In such cases, we encourage readers to identify specific constraints or tradeoffs present in their real-world situation in order to determine what tradeoffs are acceptable. For example, if the only data for a small population of a rare species consist of sparse, unverified sightings but there is a demand to protect it due to potential habitat loss, conservation practitioners will be faced with a tradeoff between action and information acquisition. We recognize that generalization simplifies across the complexity and specificity that challenges to manage and conserve rare species pose. Therefore, we do not intend for these lessons to cover all situations one may encounter managing rare species; rather, we hope that presenting our observations can serve as a resource for applied scientists to frame their thinking in complement to wildlife management and conservation training oriented around common species.

2 | LESSONS

2.1 | Standards of information must be high for rare species

When information for species is sparse, false positives (misidentifications) and false negatives (missed detections) can heavily influence data reflecting the state of a species. False positives generally receive less attention because they tend to be less important for common species. However, false positives for rare species are particularly problematic because each detection of a rare species represents a larger proportion of the total detections of the species and therefore has a greater influence on inference about the species (McKelvey et al., 2008; Figure 3). For example, the presence or absence of a single individual of a rare species can trigger a change in management or regulatory decisions (e.g., National Lynx Management Direction [USFS, 2007] or ESAs). Rare species observations are often indirect (e.g., rely on sign,

Lesson	Knowledge	Action	Inquiry
1. Standards of information must be high for rare species	•	•	•
2. Understand drivers of rarity to inform new inquiry and action	•	•	•
3. Lack of knowledge will persist for most rare species and should not always limit action, but tools from decision analyses can help	•	•	
4. Use informed assumptions when missing information	•	•	
5. Consider alternatives to trend for assessing rare species status	•		•
6. Natural history documentation is still important for rare species	•		•
7. More in-depth information about individuals in rare species populations can be more useful than a high volume of minimal depth population information			•
8. Extrapolating rare species information from a multi-species effort should be done with caution			•
9. With sparse data, simple models often offer the best inference			•
10. Curiosity-driven inquiry has the greatest potential to be transformative			•

FIGURE 2 Lessons for translating common species mindsets to work on rare species span the work of utilizing existing knowledge, engaging in inquiry to learn more and acting for conservation.

sound, or structures created by the animal), making them more prone to species misidentifications (e.g., Gooliaff & Hodges, 2018), and experts are more prone to misidentifying common species as rare (Farmer et al., 2012). As such, we need to be more confident that a rare species detection is not a misidentification of a common species through use of rigorous species verification methods (see McKelvey et al., 2008, fig. 2). Additionally, field and model methods that use multiple sources of verified data can be used to address false negatives (e.g., Golding et al., 2017; Specht et al., 2017), and false positives (Rojas et al., 2019; Ruiz-Gutierrez et al., 2016; Strickfaden et al., 2020). Recent research has also demonstrated that verified common species data from diverse

sources (e.g., citizen science) can be informative for rare species when using modern modeling methods (Crawford et al., 2020).

2.1.1 | Advice

One should determine the appropriate standard for identification of a rare species and establish a screening process for detections. Relying on unverified, anecdotal evidence for rare species, even if the volume of that evidence seems large, is potentially misleading (e.g., there are many big-foot “sightings,” but the volume does not suggest anything about its existence) (McKelvey et al., 2008).

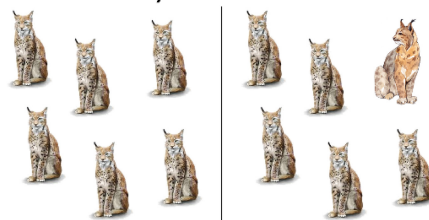
Common

Bobcat (*Lynx rufus*)

Rare

Canada lynx (*Lynx canadensis*)

Scenario 1: Lynx misidentified as bobcat

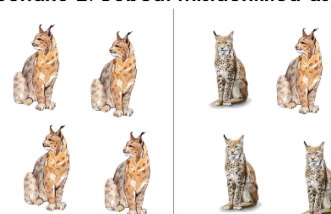


Observed True

6 bobcats 5 bobcats, 1 Canada lynx

Scenario 1 implications for common species: 20% error

Scenario 2: Bobcat misidentified as lynx



Observed True

4 Canada lynx 3 bobcats, 1 Canada lynx

Scenario 2 implications for rare species: 300% error

FIGURE 3 An example of a common species, bobcat (*Lynx rufus*), and a rare species, Canada lynx (*Lynx canadensis*) which can be misidentified as the other species where their ranges overlap (e.g., Gooliaff & Hodges, 2018) (right). Misidentification of the rare species as common (top right) has a smaller impact on our understanding of the common species. Misidentification of the common species as rare (bottom right), which has more opportunities to occur with a greater number of the common species on the landscape than the rare, will have a larger impact on understanding of the rare species. Bobcat image by Michelle Cadwell.

2.2 | Understand types and drivers of rarity to inform new inquiry and action

Living in an era of anthropogenically driven biodiversity loss has brought popular attention to the decline and rarity of many species. However, rarity is influenced by both types and drivers of rarity. Distinguishing among and between types and drivers of rarity can inform opportunities for conservation knowledge, inquiry or action about the species. Types of rarity describe how a species is rare based on relative abundance, habitat specificity, or geographic range (Rabinowitz, 1981). For example, geographic range rarity includes species with restricted geographic ranges (e.g., koalas [*Phascolarctos cinereus*]); abundance rarity includes species with low relative abundance (e.g., ocean sunfish [*Mola mola*]). Drivers of rarity, on the other hand, are current factors contributing to rarity. For example, monarch butterflies (*Danaus plexippus*), which exhibit habitat specificity rarity, are suffering a decline driven partly by ongoing loss of that habitat (Thogmartin et al., 2017), presenting opportunities for intervention via habitat restoration and protection.

Understanding the type of rarity a species of interest exhibits can come from existing knowledge (e.g., scientific literature, expert knowledge, indigenous knowledge, or other documented sources) and inform new inquiry efforts. New rare species inquiry efforts that

include field surveys to detect the species should, likewise, be shaped by considering the type of rarity because rarity underpins numerically sparse data (i.e., many zeroes) which can limit understanding when such data hinder model convergence or inference (Jeliaskov et al., 2022). Generally, the probability of detecting a rare species can be increased by surveying a broader area with fewer repeat surveys rather than a smaller area with more repeat surveys (Sanderlin et al., 2014), or with an adaptive sampling strategy (e.g., Pacifici et al., 2016; Specht et al., 2017) that facilitates more data collection around confirmed detections. However, adaptive sampling may not be feasible in all environments or result in the desired information (e.g., it may be more important to accrue more confirmed occurrence locations than to estimate occurrence with good precision). In addition, constraining sampling based on types of rarity can improve inference by reducing uninformative non-detections. If rarity is habitat-based, such as for black swifts (*Cypseloides niger*) nesting in cliff faces near waterfalls, a targeted survey of the habitat may be most informative—while surveying habitat that the species has not been documented to use may lead to discovery, it will also lead to uninformative non-detections in the dataset. If a species' rarity is due to small population size, such as wolverines (*Gulo gulo*) at the southern end of their North American range, a broad area survey using methods that favor higher likelihood of detection should be used. If a

species is geographically rare, such as North Atlantic right whales (*Eubalaena glacialis*) in the northern Atlantic Ocean, the area of interest should be constrained within that geographic range—while the North Atlantic is contiguous with the South Atlantic and Arctic Oceans, a constrained sample frame of the North Atlantic would result in a more informative understanding of where the whale occurs.

Actions to protect rare species, on the other hand, should be informed by understanding current drivers of rarity, as well as how those drivers interact with rarity type. For example, if a species is rare due to habitat loss (e.g., monarch butterfly), habitat restoration may be a more suitable strategy than a captive breeding program, which would not remove the driver of rarity. Drever et al. (2012) outline the suggestions for how rarity type, drivers, and risk tolerance (see Lesson 3) can be used to make decisions about conservation actions for rare species.

2.2.1 | Advice

When considering inquiry or action for a rare population or species, it is important to determine the existing knowledge regarding type of rarity (habitat specificity, population size, or geographic range) and drivers of rarity (e.g., habitat loss). It is important to note whether each is natural (i.e., not significantly altered by humans), anthropogenic, or driven by an interaction of the two, which is likely the case in many settings (Davidson et al., 2009). For maximizing detections using field-based observations, the following sampling strategies are effective for each type of rarity; a broad effort that maximizes detection of individuals for low abundance species; a targeted, narrowed sampling approach by habitat type for habitat specialists; and, a targeted, narrowed sampling approach by geographic range for species limited by geographic range. See Jeliaskov et al. (2022) for published guidelines on addressing rarity from sampling decisions through model inference.

2.3 | Lack of knowledge will persist for most rare species and should not always limit action, but tools from decision analyses can help

Lack of information serves as both a real and perceived barrier to action in the conservation of rare species. Information gaps can result in prioritization of survey and inventory over action through legal mandates or a sometimes baseless perception that additional information would improve management action (Litman, 2008). Information limitations can be place-specific, limiting

confidence that a broader, action-warranting pattern applies to a specific place (e.g., Danielsen et al., 2010), reflect patterns difficult to discern (e.g., the indirect effects of habitat enhancement on a rare species; Anderson & Loomis, 2006) or simply be perceived due to incomplete consideration of different forms of knowledge (e.g., Reid et al., 2021). These scenarios are more likely with rare species, where information is often sparse and patterns are challenging to discern, sometimes leading to prioritization of information gathering (e.g., inquiry) when actions would otherwise be clear. Delaying action in favor of information gathering can hamper conservation outcomes (Buxton et al., 2020), especially when the timeline for actions to be impactful is shorter than the time it takes to obtain additional information.

Decision analytic frameworks offer opportunities for synergy between action and inquiry through considering when additional information would impact action. For example, Runge et al. (2011) used an Expected Value of Information framework to weigh the likelihoods of different hypotheses explaining poor recruitment in endangered whooping cranes (*Grus americana*) and corresponding management actions. With multiple mechanistic hypotheses corresponding to the same management action, they considered whether additional research to understand the mechanism would change the action that would be pursued. Such approaches can also formally integrate risk tolerance, as it is frequently the case for rare species that the consequences of the downsides of an action (or inaction) may be of greater concern than the gains of the alternative (Bolam et al., 2019).

2.3.1 | Advice

Understanding what stressors exist for a rare species and whether there are management tools or creative solutions (outside typical constraints) to address those stressors is important for effective rare species conservation. The value of additional information should be carefully considered, including whether that information might exist in another form and whether that information would change management action and the risk tolerance for (in)action (Bolam et al., 2019). Considering multiple ways of knowing can broaden options for knowledge and action (Reid et al., 2021). In some cases, a formal decision analytic tool can be useful.

2.4 | Use informed assumptions when missing information

When the status of a rare species is uncertain, expectations for how the species will respond to a conservation

action are based on life history information, such as life span or reproductive rate, which is often unknown. In such cases, leveraging the information available and borrowing information from phylogenetically or ecologically similar species can provide a basis for moving forward with actions (Kindsvater et al., 2018). For example, the US National Marine Fisheries Service used distribution and genetic information from the rockfish genus to assess whether Puget Sound rockfish (*Sebastes emphaeus*) were likely to be distinct from other rockfish for an ESA assessment (Kelly et al., 2017) under the standard of best available evidence. Information borrowing practices build off the use of model organisms in biological sciences to understand fundamental aspects of anatomy, physiology, development and ecology. Similarly, expert opinion can contribute to the assemblage of knowledge for a species where data are lacking, with a corresponding body of literature supporting best practices for recognition of uncertainty (e.g., Stern & Humphries, 2022). Modern statistical methods offer ways to account for borrowed parameters or expert opinion in inference (e.g., use of informed Bayesian priors, imputing data, integrated models that propagate uncertainty from all data sources). Nevertheless, it is important to identify when information is borrowed and the assumptions for using that information.

2.4.1 | Advice

It can be informative to compare the available information for the rare entity of interest and the desired information that is missing. Information may be truly missing or perceived as missing. Content expertise, indigenous knowledge and other sources of knowledge can provide valuable information about the rare species. If the reason for the missing information is likely to persist, a similar species or population may provide information about the range of values of your desired information. Translating information between organisms can be effective, but it is important to state and examine assumptions that allow that translation. Frameworks like the state, identify, focus, and think framework provide a stepwise process to help practitioners identify how assumptions interact with circumstances to shape outcomes (Specht et al., 2022).

2.5 | Consider alternatives to trend for assessing rare species status

Many actions in wildlife management are based on information about changes in population status and trends. Most conservation prioritization (e.g., IUCN Red List, US ESA) relies in some way on status and trends. For many

common and harvested species, trend estimation has provided necessary insight into decline drivers, providing opportunities to act for the benefit of a species. However, accurate trend information is data intensive (Taylor & Gerrodette, 1993; Wauchope et al., 2019). Difficulties of obtaining trend for rare species are additionally complicated by stochastic dynamics that are a feature of small populations (Fauvergue et al., 2012). With small populations, changes to the reproductive or survival status of a few individuals can have an outsized effect on population size. In addition, demographic stochasticity in small populations can result in large fluctuations in relative population size, which itself can be a conservation concern for rare species, as small populations with more stochastic dynamics are more likely to go extinct (Ovas-kainen & Meerson, 2010). Furthermore, this stochasticity means that changes in populations do not necessarily indicate risk to the population in the same way that trends in a common species might. As such, due to the challenge of obtaining trend data for rare species, waiting for such information can delay critical conservation action.

Repeated assessments of non-trend population health indicators offer an alternative to population trends. Metrics such as health or condition of individuals of the limiting sex, or the degree of heterozygosity or hybridization can indicate population status. For example, repeated assessments of genetic and reproductive health in Florida panthers (*Puma concolor coryi*) indicated a need for greater gene flow and prompted introduction of new individuals (Roelke et al., 1993). These assessments can be single time-point assessments of population health indicators (e.g., population heterozygosity), which can be scaled into population metrics with processes that connect the health indicators to population dynamics (e.g., goal efficient monitoring; Golding, 2022b). Assessments of these metrics can be repeated over time to understand changes in population health status, but offer information that can be acted on in place of waiting for adequate data to decipher a trend.

2.5.1 | Advice

Populations are often best understood through metrics that vary in magnitude. If a population of interest under 250 individuals (small, vulnerable to demographic and environmental stochasticity [endangered designation, IUCN, 2012]), metrics related to individuals and reproduction, including but not limited to breeding evidence (e.g., vaquita [*Phocoena sinus*]; Rojas-Bracho et al., 2022), body condition (e.g., North Atlantic right whale [*Eubalaena glacialis*]; Corkeron et al., 2018), degree of inbreeding (e.g., Florida panther; Roelke et al., 1993), or

hybridization (e.g., west slope cutthroat trout [*Oncorhynchus clarkii lewisi*]; Corsi et al., 2013) can be most informative. If a population is between 250 and 1000 individuals (medium, buffered from demographic and environmental stochasticity, but still vulnerable [IUCN, 2012]), a population-based approach that tracks different age classes, such as an integrated population model (IPM) structure, or persistence over a larger spatio-temporal context (e.g., persistence in historical locations) may be most informative. Finally, if a population is more than 1000 individuals (larger), consider traditional trend metrics or an IPM structure. Potentially appropriate metrics should also be considered with respect to management actions available for the species of interest. Although acquiring status and trend information may be infeasible on certain time or budgetary scales, that should not negate the importance of that information long-term for conservation prioritization.

2.6 | Natural history documentation is still important for rare species

The lack of basic information on a species, such as distribution or life history characteristics, can act as a barrier to conservation and management. For example, endangered Bahama orioles (*Icterus northropi*), with a few hundred individuals remaining, were thought to nest only in palm trees. However, new documentation of breeding in pine forests prompted new surveys resulting in a 10-fold increase in the population estimate (Rowley et al., 2021). In extreme cases, natural history observations have led to the rediscovery of a presumed extinct species, facilitating conservation efforts (e.g., Scheffers et al., 2011). Documenting information reduces barriers to conservation action (Tosa et al., 2021). Observations can be submitted to state natural heritage programs or observational databases such as eBird, eMammal, NatureServe, or iNaturalist. In addition to serving as databases, such platforms typically protect fine-scale location information to guard against location-driven threats (e.g., poaching, collecting, disturbance; Lindenmayer & Scheele, 2017). Natural history information collected should be published in an accessible, permanent form, such as a journal article, report or database (e.g., Conservation Evidence; Sutherland et al., 2019), to allow others to build upon it.

2.6.1 | Advice

Whenever possible, rare species information should be shared by submitting data to appropriate databases and

publishing in peer-reviewed journals and evidence platforms. In almost all platforms, information flow is two-way: contributing data and accessing other data are both important actions.

2.7 | More in-depth information about individuals of rare species can be more useful than a high volume of minimal depth population information

Each individual of a rare species contributes more information to what is known about a species than an individual of a common species. For example, in a population of 10 individuals, one individual represents 10% of the information about the population. Equally, an individual in a population of 100 is 1% of the total population, and 0.1% in a population of 1000. In large populations, patterns across individuals can be examined with averages acquired over many individuals, making generalities possible, whereas patterns across 10 individuals are more like a collection of individual characteristics, with few discernible patterns. As such, it is often advantageous to gather more information for a rare species than location and time of a detection during a survey. Consider whether information about sex, movement, reproductive status, age (e.g., Tella et al., 2013), health (e.g., Hansen et al., 2016), or heterozygosity/haplotype (e.g., Tucker et al., 2014) might assist in understanding management opportunities for a species. For example, camera trap and track plate surveys for Humboldt marten (*Martes caurina humboldtensis*) yielded detections facilitating inference about current range. However, the addition of hair snares and telemetry provided detailed information about the population (number of individuals) and individual habitat use for status assessment (Moriarty et al., 2016). Furthermore, in surveys, it may be important to expend field effort to maximize cumulative detection at a site and minimize detection uncertainty rather than optimize survey design for estimation precision and accept greater uncertainty in occurrence or abundance status of any individual site (Latif et al., 2017; Sanderlin et al., 2014).

2.7.1 | Advice

Coarse estimates of the size of a population of interest should inform rare species inquiry. Even an order of magnitude estimate (e.g., tens, hundreds, or thousands) provides a good starting point to think about whether information beyond occurrence patterns might be relevant to an endeavor (i.e., what proportion of the

information does one individual represent?). Populations with tens of individuals (below 100) may be good candidates for collecting more individual information because of the high proportion of information an individual represents.

2.8 | Extrapolating rare species information from a multispecies effort should be done with caution ⓘ

With advances in population and distribution modeling, multispecies survey efforts can produce estimates for many parameters, including those with sparse data (i.e., rare species) and such surveys can provide valuable opportunities to infer rare species dynamics. However, analyses of rare species data from a multispecies effort, including raw detection results or model estimates, should be interpreted with a comparison of the information goals for the rare species relative to the survey design. Model estimates from a multispecies effort may be generated automatically if the analysis is inclusive of all surveyed species, even if a species is undetected in a particular sampling time or place. For example, the Integrated Monitoring in Bird Conservation Regions program generates occupancy estimates for American avocets (*Recurvirostra americana*) in Oregon even in years where there is only one detection (Rocky Mountain Bird Observatory, 2023). When a species is infrequently detected, model estimates rely heavily on other information in the model (e.g., priors, information from the same species in other places or times, information from similar species), with estimates therefore reflecting assumptions that the species of interest shares temporal, spatial or life history similarity with the groups whose data are supporting those estimates. These sparse data estimates may be useful in some cases but can be inappropriate to base spatially specific, single species inference on. Recent research in species distribution modeling has shown that careful selection of multispecies community data improves rare species occurrence predictions, although is unlikely to improve abundance predictions (Zhang et al., 2020). Further, spatial predictions about rare species occurrence from multispecies models should be examined, as the assumptions may not necessarily apply well to all rare species. For example, a random sample of the landscape may provide a good basis for understanding the distribution of a species that uses a widespread habitat type (e.g., northern bog lemming [*Synaptomys borealis*], in the tundra), but not for one with high habitat specificity (e.g., northern bog lemming at the southern extent of its range where it persists in sparsely distributed fen wetlands).

2.8.1 | Advice

Multispecies efforts are often designed for a specific group of species that may or may not include a rare species of interest. Therefore, it is important to consider if information goals for the rare species are met by the multispecies effort, or if a targeted effort would be appropriate. If the information goals for the rare species are met, one should also consider what proportion of surveys the species was detected in. If it is <1% the sample is likely too small to make reliable inference about the rare species. If it is >20% it may provide a reliable estimate. If it is between 1 and 20%, estimates for the rare species may reflect the mean of the community. This may reflect the lack of data for the rare species rather than the actual pattern in the rare species. If rare species might respond differently from the species in the community, an additional effort targeted for the rare species may provide better information. Finally, the spatial scale of inference for the multispecies effort may be inappropriate for the rare species.

2.9 | With sparse data, simple models often offer the best inference ⓘ

Rare species modeling is often conducted with sparse data. One of the problems with producing models with sparse data is model overfitting, where we strive to represent too much complexity through the inclusion of many model variables, while limited by data too sparse to inform such models (Breiner et al., 2015; Giudice et al., 2012). Although strategies have been proposed to overcome this problem, including ensembles of small models (Breiner et al., 2015), one straightforward approach is a simplified model; well-informed, limited inference can be better than ill-informed, over-extended inference (Starfield & Beloch, 1986). For example, using a composite habitat suitability model as a covariate when assessing a species' response to habitat impacts could allow inclusion of habitat suitability information while expending only a single degree of freedom (Specht, 2018). Simpler models with sparse data may limit inference, but when they promote better decisions than could be made without them, they have proven their utility (Millsbaugh et al., 2009). Furthermore, evidence suggests that reducing uncertainty and improving prediction does not make information more likely to be used in conservation, particularly for species distribution models (Tulloch et al., 2016).

2.9.1 | Advice

Rare species models can be simplified with procedures such as degrees of freedom spending (outlined in Giudice

et al., 2012, appendix A). In addition, variable construction may be limiting the models available; there may be opportunities for grouping, recategorizing, or changing the representation of the information (i.e., convert to a continuous variable) that can aid in inference.

2.10 | Curiosity-driven inquiry has some of the greatest potential to be transformative ①

When a field of inquiry is unexplored, curiosity can be one of the main drivers of advances. Many ecologists can relate to a sense of curiosity that seems innate. In fact, psychologists have identified this as epistemic curiosity (Berlyne, 1954) and shown that curiosity that drives new discovery elicits feelings of enjoyment (Litman, 2008). The combination of epistemic curiosity combined with the fact that people value rare species (Angulo et al., 2009; Angulo & Courchamp, 2009; Gaston, 1994), means that people will study rare species simply because they find joy in the process. Although inquiry without direct application can be difficult to fund in conservation, it can lead to major advances in conservation. For example, curiosity-driven marking of individual animals has underpinned innumerable conservation actions resulting from improved understanding of movement and reproductive ecology (e.g., Bairlein, 2001). Similarly, basic physiological research, such as quantifying the physiological reproductive problems that accumulate over a lifespan due to stressful experiences, is already being applied to developing methods to improve wildlife reproduction in conservation (Comizoli & Holt, 2019). The path from the unknown to conservation action can be long, but it is shaped by disruptions to fundamental understandings. Sometimes pursuing knowledge due to curiosity is the linchpin needed for an advance in rare species conservation.

2.10.1 | Advice

Novel findings can be transformative in conservation, where action is often based upon less information than would be ideal. Opportunities that “outside the box” inquiries offer may improve science and conservation for rare species.

3 | CONCLUSIONS

Wildlife conservation management is extending to a breadth of species not yet on the brink of extinction and ushering in a new era for conservation. Those involved in information-to-action roles of wildlife conservation

management will need to adapt tools and knowledge to the challenges presented by rare species. The guidance presented in this paper offers ideas for adapting information-to-action skills from common species to the different challenges presented by rare species. The considerations present in an information-to-action pathway for any species involve weighing tradeoffs, and the ideas presented herein are no different in that regard. For example, guidance from lessons 1 (“Standards of information must be high for rare species”) and 3 (“Lack of knowledge will persist for most rare species and therefore should not always limit action”), if considered independent of a specific case, may appear to conflict. However, in the context of a specific case, consideration of these same lessons may prompt simultaneous strategies for action and inquiry, or a decision about whether a case warrants action (absent desired information) or inquiry to obtain that information. As we grow the toolbox applicable to the conservation management of rare species, we call for such approaches to be incorporated into the training of wildlife professionals.

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CONFLICT OF INTEREST STATEMENT

The authors declare they have no conflict of interest.

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REFERENCES

- Anderson, L. E., & Loomis, D. K. (2006). Balancing stakeholders with an imbalanced budget: How continued inequities in wildlife funding maintains old management styles. *Human Dimensions of Wildlife*, 11, 455–458.
- Angulo, E., & Courchamp, F. (2009). Rare species are valued big time. *PLoS One*, 4, e5215.
- Angulo, E., Deves, A. L., Saint Jalmes, M., & Courchamp, F. (2009). Fatal attraction: Rare species in the spotlight. *Proceedings of the Royal Society B: Biological Sciences*, 276, 1331–1337.
- Australian Government Department of Agriculture, Fisheries, and Forestry. (2021). Threatened species strategy action plan 2021–2026.
- Bairlein, F. (2001). Results of bird ringing in the study of migration routes and behaviour. *Ardea*, 89, 7–19.
- Berlyne, D. E. (1954). A theory of human curiosity. *British Journal of Psychology*, 45, 180–191.
- Bolam, F. C., Grainger, M. J., Mengersen, K. L., Stewart, G. B., Sutherland, W. J., Runge, M. C., & McGowan, P. J. (2019). Using the value of information to improve conservation decision making. *Biological Reviews*, 94, 629–647.

- Borgelt, J., Dorber, M., Høiberg, M. A., & Verones, F. (2022). More than half of data deficient species predicted to be threatened by extinction. *Communications Biology*, 5, 679.
- Breiner, F. T., Guisan, A., Bergamini, A., & Nobis, M. P. (2015). Overcoming limitations of modelling rare species by using ensembles of small models. *Methods in Ecology and Evolution*, 6, 1210–1218.
- Brulle, R. J., & Benford, R. D. (2012). From game protection to wildlife management: Frame shifts, organizational development, and field practices. *Rural Sociology*, 77, 62–88.
- Buxton, R. T., Avery-Gomm, S., Lin, H.-Y., Smith, P. A., Cooke, S. J., & Bennett, J. R. (2020). Half of resources in threatened species conservation plans are allocated to research and monitoring. *Nature Communications*, 11, 4668.
- Comizzoli, P., & Holt, W. V. (2019). Breakthroughs and new horizons in reproductive biology of rare and endangered animal species. *Biology of Reproduction*, 101, 514–525.
- Convention on Biological Diversity (CBD). (2022). Kunming-Montreal global biodiversity framework.
- Corkeron, P., Hamilton, P., Bannister, J., Best, P., Charlton, C., Groch, K. R., Findlay, K., Rowntree, V., Vermeulen, E., & Pace, R. M. (2018). The recovery of north Atlantic right whales, *Eubalaena glacialis*, has been constrained by human-caused mortality. *Royal Society Open Science*, 5, 180892.
- Corsi, M. P., Eby, L. A., & Barfoot, C. A. (2013). Hybridization with rainbow trout alters life history traits of native westslope cutthroat trout. *Canadian Journal of Fisheries and Aquatic Sciences*, 70, 895–904.
- Crawford, B. A., Olds, M. J., Maerz, J. C., & Moore, C. T. (2020). Estimating population persistence for at-risk species using citizen science data. *Biological Conservation*, 243, 108498.
- Crisfield, V. E., Guillaume Blanchet, F., Raudsepp-Hearne, C., & Gravel, D. (2024). How and why species are rare: Towards an understanding of the ecological causes of rarity. *Ecography*, 2024, e07037.
- Danielsen, F., Burgess, N. D., Jensen, P. M., & Pirhofer-Walzl, K. (2010). Environmental monitoring: The scale and speed of implementation varies according to the degree of peoples involvement. *Journal of Applied Ecology*, 47, 1166–1168.
- Davidson, A. D., Hamilton, M. J., Boyer, A. G., Brown, J. H., & Ceballos, G. (2009). Multiple ecological pathways to extinction in mammals. *Proceedings of the National Academy of Sciences of the United States of America*, 106, 10702–10705.
- Drever, C. R., Drever, C. M., & Sleep, J. H. (2012). Understanding rarity: A review of recent conceptual advances and implications for conservation of rare species. *The Forestry Chronicle*, 88, 165–175.
- Farmer, R. G., Leonard, M. L., & Horn, A. G. (2012). Observer effects and avian-call-count survey quality: Rare-species biases and overconfidence. *The Auk*, 129, 76–86.
- Fauvergue, X., Vercken, E., Malausa, T., & Hufbauer, R. A. (2012). The biology of small, introduced populations, with special reference to biological control. *Evolutionary Applications*, 5, 424–443.
- Feldpausch-Parker, A. M., Parker, I. D., & Vidon, E. S. (2017). Privileging consumptive use: A critique of ideology, power, and discourse in the North American model of wildlife conservation. *Conservation and Society*, 15, 33–40.
- Gaston, K. J. (1994). What is rarity? In K. J. Gaston (Ed.), *Rarity* (pp. 1–21). Springer.
- Gaston, K. J., Blackburn, T. M., Greenwood, J. J., Gregory, R. D., Quinn, R. M., & Lawton, J. H. (2000). Abundance–occupancy relationships. *Journal of Applied Ecology*, 37, 39–59.
- Gaston, K. J., & Fuller, R. A. (2008). Commonness, population depletion and conservation biology. *Trends in Ecology & Evolution*, 23, 14–19.
- Giudice, J. H., Fieberg, J. R., & Lenarz, M. S. (2012). Spending degrees of freedom in a poor economy: A case study of building a sightability model for moose in northeastern Minnesota. *Journal of Wildlife Management*, 76, 75–87.
- Golding, J. D. (2022a). Use of terms rare species, endangered species, threatened species, and listed species in the Journal of Wildlife Management from 1937–2022 (Dataset).
- Golding, J. D. (2022b). *Rethinking rare: Novel approaches to rare species monitoring and conservation*. University of Montana.
- Golding, J. D., Nowak, J. J., & Dreitz, V. J. (2017). A multispecies dependent double-observer model: A new method for estimating multispecies abundance. *Ecology and Evolution*, 7, 3425–3435.
- Gooliaff, T., & Hodges, K. E. (2018). Measuring agreement among experts in classifying camera images of similar species. *Ecology and Evolution*, 8, 11009–11021.
- Handmaker, O., Keeler, B. L., & Milz, D. (2021). What type of value information is most valuable to stakeholders? Multi-sector perspectives on the utility and relevance of water valuation information. *Environmental Science & Policy*, 115, 47–60.
- Hansen, W. K., Bate, L. J., Landry, D. W., Chastel, O., Parenteau, C., & Breuner, C. W. (2016). Feather and faecal corticosterone concentrations predict future reproductive decisions in harlequin ducks (*Histrionicus histrionicus*). *Conservation Physiology*, 4, cow015.
- IUCN (Ed.). (2012). *IUCN Red List categories and criteria: Version 3.1* (2nd ed.). IUCN.
- Jeliazkov, A., Gavish, Y., Marsh, C. J., Geschke, J., Brummitt, N., Rocchini, D., Haase, P., Kunin, W. E., & Henle, K. (2022). Sampling and modelling rare species: Conceptual guidelines for the neglected majority. *Global Change Biology*, 28, 3754–3777.
- Kelly, R. P., Levin, P. S., & Lee, K. N. (2017). Science, policy, and data-driven decisions in a data vacuum. *Ecology Law Quarterly*, 44, 7–40.
- Kindsvater, H. K., Dulvy, N. K., Horswill, C., Juan-Jordá, M.-J., Mangel, M., & Matthiopoulos, J. (2018). Overcoming the data crisis in biodiversity conservation. *Trends in Ecology & Evolution*, 33, 676–688.
- Latif, Q. S., Ellis, M. M., Saab, V. A., & Mellen-McLean, K. (2017). Simulations inform design of regional occupancy-based monitoring for a sparsely distributed, territorial species. *Ecology and Evolution*, 8, 1171–1185.
- Le Viol, I., Jiguet, F., Brotons, L., Herrando, S., Lindström, Å., Pearce-Higgins, J. W., Reif, J., Turnhout, C. V., & Devictor, V. (2012). More and more generalists: Two decades of changes in the European avifauna. *Biology Letters*, 8(5), 780–782.
- Lindenmayer, D., & Scheele, B. (2017). Do not publish. *Science*, 356, 800–801.
- Litman, J. A. (2008). Interest and deprivation factors of epistemic curiosity. *Personality and Individual Differences*, 44, 1585–1595.
- McKelvey, K. S., Aubry, K. B., & Schwartz, M. K. (2008). Using anecdotal occurrence data for rare or elusive species: The illusion of reality and a call for evidentiary standards. *Bioscience*, 58, 549–555.

- Meretsky, V., Maguire, L. A., Davis, F. W., Stoms, D. M., Scott, J. M., Figg, D., Goble, D. D., Griffith, B., Henke, S. E., Vaughn, J., & Yaffee, S. L. (2012). A state-based national network for effective wildlife conservation. *Bioscience*, 62(11), 970–976.
- Millspaugh, J. J., Gitzen, R. A., Larsen, D. R., Larson, M. A., & Thompson, F. R. (2009). General principles for developing landscape models for wildlife conservation. In J. J. Millspaugh & F. R. Thompson (Eds.), *Models for planning wildlife conservation in large landscapes* (pp. 1–31). Elsevier.
- Moriarty, K. M., Bailey, J. D., Smythe, S. E., & Verschuy, J. (2016). Distribution of Pacific marten in coastal Oregon. *Northwestern Naturalist*, 97(2), 71–81.
- Ovaskainen, O., & Meerson, B. (2010). Stochastic models of population extinction. *Trends in Ecology & Evolution*, 25, 643–652.
- Pacifici, K., Reich, B. J., Dorazio, R. M., & Conroy, M. J. (2016). Occupancy estimation for rare species using a spatially-adaptive sampling design. *Methods in Ecology and Evolution*, 7, 285–293.
- Rabinowitz, D. (1981). Seven forms of rarity. In H. Synge (Ed.), *The biological aspects of rare plant conservation* (pp. 205–217). Wiley & Sons.
- Reid, A. J., Eckert, L. E., Lane, J. F., Young, N., Hinch, S. G., Darimont, C. T., Cooke, S. J., Ban, N. C., & Marshall, A. (2021). “Two-Eyed Seeing”: An Indigenous framework to transform fisheries research and management. *Fish and Fisheries*, 22, 243–261.
- Rocky Mountain Bird Observatory. (2023). Rocky Mountain Avian Data Center [database]. Retrieved from <https://rmbo.org/v3/avian/ExploretheData.aspx>
- Roelke, M. E., Martenson, J. S., & O'Brien, S. J. (1993). The consequences of demographic reduction and genetic depletion in the endangered Florida panther. *Current Biology*, 3, 340–350.
- Rojas, V. G., Loeb, S. C., & O'Keefe, J. M. (2019). False-positive occupancy models produce less-biased occupancy estimates for a rare and elusive bat species. *Journal of Mammalogy*, 100, 212–222.
- Rojas-Bracho, L., Taylor, B., Booth, C., Thomas, L., Jaramillo-Legorreta, A., Nieto-García, E., Hinojosa, G. C., Barlow, J., Mesnick, S. L., Gerrodette, T., Olson, P., Henry, A., Rizo, H., Hidalgo-Pla, E., & Bonilla-Garzón, A. (2022). More vaquita porpoises survive than expected. *Endangered Species Research*, 48, 225–234.
- Rowley, M., Stanley, R., Antalffy, J., Christilf, J., Stonko, D., Johnson, S., Cant-Woodside, S., Sillett, T., Fagan, M., Studds, C., & Omland, K. (2021). Hierarchical distance sampling reveals increased population size and broader habitat use in the endangered Bahama Oriole. *Avian Conservation and Ecology*, 16(1), 5.
- Ruiz-Gutierrez, V., Hooten, M. B., & Campbell Grant, E. H. (2016). Uncertainty in biological monitoring: A framework for data collection and analysis to account for multiple sources of sampling bias. *Methods in Ecology and Evolution*, 7, 900–909.
- Runge, M. C., Converse, S. J., & Lyons, J. E. (2011). Which uncertainty? Using expert elicitation and expected value of information to design an adaptive program. *Biological Conservation*, 144, 1214–1223.
- Sanderlin, J. S., Block, W. M., & Ganey, J. L. (2014). Optimizing study design for multi-species avian monitoring programmes. *Journal of Applied Ecology*, 51, 860–870.
- Scheffers, B. R., Yong, D. L., Harris, J. B. C., Giam, X., & Sodhi, N. S. (2011). The world's rediscovered species: Back from the brink? *PLoS One*, 6, e22531.
- Specht, H. (2018). *Habitat use and reproductive success of waterbirds in the human-dominated landscape of North America's prairies: Using sparse data to inform management*. Dissertation, University of Minnesota.
- Specht, H. M., Golding, J. D., Pero, E. M., Crane, M. T., Ortiz-Calo, W., McDevitt, M. C., Karlen, J. G., Branch, J. V., Hansen, C. P., & Millspaugh, J. J. (2022). A framework for evaluating the implications of assumptions in broad conservation strategies. *Conservation Science and Practice*, 4, e12758.
- Specht, H. M., Reich, H. T., Iannarilli, F., Edwards, M. R., Stapleton, S. P., Weegman, M. D., Johnson, M. K., Yohannes, B. J., & Arnold, T. W. (2017). Occupancy surveys with conditional replicates: An alternative sampling design for rare species. *Methods in Ecology and Evolution*, 8, 1725–1734.
- Starfield, A. M., & Beloch, A. L. (1986). *Building models for conservation and wildlife management*. Macmillan Publishing Company.
- Stern, E. R., & Humphries, M. M. (2022). Interweaving local, expert, and indigenous knowledge into quantitative wildlife analyses: A systematic review. *Biological Conservation*, 266, 109444.
- Strickfaden, K. M., Fagre, D. A., Golding, J. D., Harrington, A. H., Reintsma, K. M., Tack, J. D., & Dreitz, V. J. (2020). Dependent double-observer method reduces false-positive errors in auditory avian survey data. *Ecological Applications*, 30, e02026.
- Sutherland, W. J., Taylor, N. G., MacFarlane, D., Amano, T., Christie, A. P., Dicks, L. V., Lemasson, A. J., Littlewood, N. A., Martin, P. A., Ockendon, N., Petrovan, S. O., Robertson, R. J., Rocha, R., Shackelford, G. E., Smith, R. K., Tyler, E. H. M., & Wordley, C. F. R. (2019). Building a tool to overcome barriers in research-implementation spaces: The conservation evidence database. *Biological Conservation*, 238, 108199.
- Taylor, B. L., & Gerrodette, T. (1993). The uses of statistical power in conservation biology: The vaquita and northern spotted owl. *Conservation Biology*, 7, 489–500.
- Teel, T. L., Bruyere, B., Dayer, A., Stoner, K. E., Bishop, C., Bruskotter, J., Freeman, S., Newmark, J., Jager, C., & Manfredo, M. J. (2022). Reenvisioning the university education needs of wildlife conservation professionals in the United States. *Conservation Science and Practice*, 4, e610.
- Tella, J. L., Rojas, A., Carrete, M., & Hiraldo, F. (2013). Simple assessments of age and spatial population structure can aid conservation of poorly known species. *Biological Conservation*, 167, 425–434.
- Thogmartin, W. E., Wiederholt, R., Oberhauser, K., Drum, R. G., Diffendorfer, J. E., Altizer, S., Taylor, O. R., Pleasants, J., Semmens, D., Semmens, B., Erickson, R., Libby, K., & Lopez-Hoffman, L. (2017). Monarch butterfly population decline in North America: Identifying the threatening processes. *Royal Society Open Science*, 4, 170760.
- Thompson, W. (Ed.). (2004). *Sampling rare or elusive species: Concepts, designs, and techniques for estimating population parameters*. Island Press.
- Tosa, M. I., Dziedzic, E. H., Appel, C. L., Urbina, J., Massey, A., Ruprecht, J., Eriksson, C. E., Dolliver, J. E., Lesmeister, D. B., Betts, M. G., Peres, C. A., & Levi, T. (2021). The rapid rise of next-generation natural history. *Frontiers in Ecology and Evolution*, 9, 698131.
- Tucker, J. M., Schwartz, M. K., Truex, R. L., Wisely, S. M., & Allendorf, F. W. (2014). Sampling affects the detection of genetic subdivision and conservation implications for fisher in the Sierra Nevada. *Conservation Genetics*, 15, 123–136.

- Tulloch, A. I. T., Sutcliffe, P., Naujokaitis-Lewis, I., Tingley, R., Brotons, L., Ferraz, K. M. P. M. B., Possingham, H., Guisan, A., & Rhodes, J. R. (2016). Conservation planners tend to ignore improved accuracy of modelled species distributions to focus on multiple threats and ecological processes. *Biological Conservation*, 199, 157–171.
- USDA Forest Service (USFS). (2007). *Northern Rockies lynx management direction: Record of decision*. U.S. Department of Agriculture, Forest Service.
- Wauchope, H. S., Amano, T., Sutherland, W. J., & Johnston, A. (2019). When can we trust population trends? A method for quantifying the effects of sampling interval and duration. *Methods in Ecology and Evolution*, 10, 2067–2078.
- Zhang, C., Chen, Y., Xu, B., Xue, Y., & Ren, Y. (2020). Improving prediction of rare species' distribution from community data. *Scientific Reports*, 10, 12230.

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